

Marketing Campaign Response — Summary

Predicting who responds to a campaign, and turning it into a targeting and budget plan.

► **Live interactive app:** <https://marketing-campaign-response-ajrjgsucchbawv2ykmcklt.streamlit.app>

📄 **Full Jupyter notebook:** [view the complete analysis & code](#)

2,237
CUSTOMERS

14.9%
BASELINE
RESPONSE

XGBoost
BEST MODEL

0.90
ROC-AUC

~3x
WARM-LIST LIFT

Bottom line. A retailer's last campaign reached everyone, but only ~15% responded, so most of the spend was wasted. The model scores each customer's likelihood to respond (ROC-AUC $\approx$ 0.90). Aimed at the right people, the same budget earns far more.

Approach

Cleaned 2,237 customer records (filled 24 missing incomes, removed a few typo rows, tidied messy labels), then built 10 behaviour features: spending, recency, engagement, past-campaign history, income tiers. Trained and compared three models inside leak-free scikit-learn pipelines with 5-fold cross-validation. Because only ~15% respond, models are judged on ranking quality (ROC-AUC) and recall, not plain accuracy, and each is told to weight the rare "yes" cases.

Key insights

- **Past responders are gold:** people who accepted a past campaign respond at **41%** vs **8%** (~3x).
- **Higher income, higher response:** **28%** vs **10%**; high income + a past acceptance reaches **47%**.
- **Recent buyers are warm:** responders last bought ~35 days ago vs ~52 for everyone else.
- **Households without kids respond more:** **27%** vs **10%** with children.
- **Catalog buyers over-respond** (4.2 vs 2.4 purchases); website *visits* don't predict response at all.

What drives a response

The model's top factors, in order: **past campaign acceptances** (by far the strongest), the **recency** of the last purchase, **customer tenure**, then **income and spending** (especially wine, meat, and catalog buying). Being single or highly educated nudges response up; children at home pull it down. In one sentence: the most likely responder is a **recent, comfortable, previously responsive buyer with no kids at home**, and the model puts a number on each trait.

Model performance

Model	Acc	Prec	Rec	F1	AUC
XGBoost ★	0.89	0.63	0.61	0.62	0.90
Random Forest	0.89	0.72	0.42	0.53	0.89
Logistic Reg.	0.81	0.42	0.73	0.53	0.88

XGBoost wins on ranking (ROC-AUC) and F1. Since a missed responder (~\$11) costs more than a wasted contact (~\$3), we lean toward recall and set the contact cutoff where expected profit is highest, not at a textbook 0.5.

Business recommendations

- **Contact first:** rank everyone by predicted probability; start with the ~460 prior responders (~3x the base rate, free to find).
- **Budget:** concentrate on the top 2–3 probability deciles; stop funding the unresponsive bottom half.
- **Personalise:** premium offers for high-income responders, recency-triggered sends for warm buyers, sustained catalog investment.
- **Prove it:** track profit per contact against a random-targeting control so the lift is provable in dollars.

Deployment risks & what's next

Watch for: wasted spend from false positives (cap contact frequency, tune the cutoff), missed sales from false negatives (lean toward recall, keep a small random holdout), and data drift as customer behaviour shifts (monitor the inputs, retrain on a rolling quarterly window). **Next steps:** customer segmentation to pair the right offer with each group, an A/B test against random targeting to prove the lift in dollars, and probability calibration so the scores can be trusted as money.

Built with Python, pandas, scikit-learn, XGBoost, and Streamlit. The full method, code, and charts live in the accompanying Jupyter notebook (linked above).